MIL-B-44054B(GL)
18 June 1985
SUPERSEDING
MIL-B-44054A(GL)
29 March 1983

MILITARY SPECIFICATION

BOTTLE: OIL SAMPLE; WITH CONTAINER, NONAERONAUTICAL

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers oil sample bottles with containers.

1.2 Classification. The bottles with containers shall be of the following types as specified (see 6.2):

Type I - Twelve 3-ounce bottles with caps and container

Type II - Two 3-ounce bottles with caps, two plastic bags and container

Type III - One 3-ounce bottle with cap, one plastic bag and one shipping sack (container)

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin

PPP-B-26 - Bag, Plastic (General Purpose)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8125

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PPP-B-566 - Boxes, Folding, Paperboard

PPP-B-636 - Boxes, Shipping, Fiberboard

PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

PPP-S-30 - Sacks, Shipping, Paper (Cushioned)

MILITARY

MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

PLASTIC BOTTLE INSTITUTE (PBI)

Division of the Society of Plastics Industry

DRAWING SP - 400

(Application for copies should be addressed to the Plastic Bottle Institute, 355 Lexington Ave., New York, NY 10017)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Materials and components (see 6.3).

3.1.1 Containers.

3.1.1.1 Type I - Shipping box. The shipping box shall be fabricated from corrugated fiberboard material conforming to type CF, class domestic, grade 275 of PPP-F-320. The box shall be constructed in accordance with style CSSC of PPP-B-636. The manufacturer's joint shall be glued. The bottom flaps shall be closed in accordance with method I or II of PPP-B-636. If staples are used for closure of the bottom flaps, the box shall be provided with a bottom pad made of the same material as the box. Alternatively, box style may be a one-piece folder with top flap tuck and center tuck bottom, or a die-cut, self-erecting folder having a single or double thickness bottom, triple thickness ends and single thickness back and top with a tuck top closure. Each box shall be provided with full-height chipboard partitions that divide the interior of the box into twelve equal sized cells. The box size shall be such that the capped bottles fit into the cells without bulging the partitions nor allow more than 1/8 inch clearance between the bottles and the surfaces of the partition or the box.

3.1.1.2 Type II - Mailing carton. The carton shall be made from paperboard, 5 inches in length, 2 inches in width and 3 inches in depth conforming to variety 1, style II, type D, class A, tongue and tuck of PPP-B-566.

3.1.1.3 Type III - Shipping sack. The shipping sack shall be 6 inches by 10 inches conforming to type I or III of PPP-S-30.

3.1.2 Shipping box marking. Each type I shipping box shall be marked on two sides with arrows pointing to the top of the box to indicate the position in which the box shall be handled during transit and storage. The word "UP" shall be placed beside the arrows. The letters and arrows shall be at least 1-1/2 inches in height.

* 3.1.3 Oil sample bottle. The oil sample bottle shall be a wide-mouth cylindrical bottle of one piece construction fabricated of natural, translucent, high density polyethylene with a flame treatment that permits printing ink to adhere. The bottle shall have a threaded finish on top and shall have a rounded shoulder below the bottom of the neck. The open end of

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the bottle shall have the following characteristics to accommodate a screwcap closure (see 3.1.5).

a. Continuous type exterior thread in accordance with L style thread profile of Drawing No. SP-400.

b. Flat top surface at the opening that is level and smooth so as to create a leak proof seal with the cap liner.

The bottle shall have a cylindrical shape and shall remain in an upright position when placed on a flat and level surface. The fill capacity of the bottle shall be 95 ± 15 mL when tested as specified in 4.3.1 The exterior surfaces of the bottle shall be smooth, except that impressions from commercial markings, including index slot, and seams from mold marks shall be acceptable but free of flash. The bottle shall be free of cracks, chips, breaks, deformation, sharp edges and burrs. The bottle shall conform to the dimensional requirements specified in table I.

* TABLE I. Bottle dimensions

Inner diameter of opening (mouth and neck)	1.405 inches ± .03 inch
Overall height (with screwcap closure in place)	3.250 inches max.
Wall thickness	0.015 inch min.
Top surface thickness (Brim)	0.060 inch min.
Overall width	2.0 inches max.
Threads	43/400 mm

* 3.1.4 Bottle marking. Each bottle shall be printed and blocked as shown on figure 1. Markings shall be applied directly on the bottle by screen printing or any other commercial means using ink that is impervious to oil, diesel fuel, and water. The ink for the printed and blocked areas as shown on figure 1 shall be white and permanently adhered to the bottle. The blocked areas shall be capable of being written on with either ball point pen or pencil. The writing shall remain legible after testing as specified in 4.3.4.

* 3.1.5 Screwcap. The screwcap shall be a one-piece metal cap of rigid construction having inner edges sufficiently threaded with a continuous type thread to permit closure on the bottle (see 3.1.3). A liner shall be secured to the inside of the cap in such a manner that it does not fall out of the cap, or is not off-center when in the closed position on the bottle. The liner shall be supplied with a facing that is impervious to water, diesel fuel, and oil. The liner shall be of sufficient thickness and resilience to

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form an effective seal with the flat top surface of the bottle (see 3.1.3). The cap shall not crack, split, break or leak when tested as specified in 4.2.1.4. The cap shall be free of deformation, cracks, sharp edges, burrs, or protrusions. The cap shall have a finish to resist corrosion.

3.1.6 Bag, plastic. The plastic bags shall conform to type II, style 3 of PPP-B-26 with inside dimensions of 6 inches by 6 inches. The plastic film shall conform to type I, class 1, grade A, finish 1, single wall, 0.004 inch thick of L-P-378.

3.2 Container assembly.

3.2.1 Bottle. Each bottle shall have the screwcap secured in place finger tight.

3.2.2 Type I. When applicable (see 3.1.1.1), a fiberboard pad shall be placed in the bottom of the shipping box. The chipboard partitions shall be set up within the box and twelve bottles, with screw caps secured, shall be placed in each box. The top flaps of CSSC style boxes shall be loosely closed. Alternative style boxes shall be closed by means of the top flap tuck or the tuck top closure, as applicable.

3.2.3 Type II. Two capped bottles shall be placed in the mailing carton. The two plastic bags shall be neatly folded and placed between the two bottles. Both ends of the carton shall be closed and tucked.

3.3 Workmanship. The finished oil sample bottles with container shall conform to the quality of product established by this document. The bottles, caps, containers, and bags (when applicable) shall be free from defects which detract from their appearance or may impair their serviceability. All components shall be clean, dry and free of dirt or foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the

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requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. In addition, the bottles, caps, and boxes shall be examined in accordance with 4.2.1.1 through 4.2.1.3. Bottles, with caps, shall be tested for the characteristics specified in table II. The lot size shall be expressed in units of bottles with caps. The sample unit shall be one bottle with cap. The inspection level shall be S-1 and the acceptable quality level (AQL) expressed in terms of defects per hundred units, shall be 2.5.

TABLE II - Component tests

Characteristic	Requirement reference	Test method
Capacity	3.1.3	4.3.1
Leakage	3.1.3	4.3.2
Pen and pencil marking capability	3.1.4	4.3.3
Legibility after marking	3.1.4	4.3.4

* 4.2.1.1 Bottle and cap examination. The bottles and caps shall be examined for the defects listed below. The lot size shall be expressed in units of bottles with caps. The sample unit shall be one bottle with cap. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for total (major and minor combined) defects.

Defect	Classification Major	Minor
Bottle or cap not free of cracks, chips, or breaks	X	
Bottle or cap not free of distortion	X	
Bottle or cap not free of sharp edges or burrs	X	
Top of bottle not flat or smooth	X	
Bottle not clean, dry, or ready for use	X	
Bottle not free of foreign matter	X	

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Defect	Classification Major	Minor
NOTE: Application of cap to bottle during capping may result in deposit of some small plastic or fiber shavings in the bottle. Presence of a few of these shavings shall not be considered a defect.		
Liner not aligned or secured within screw cap	X	
Bottle not stable when placed on a flat and level surface	X	
Thread on cap or bottle not free of stripping, distortion or discontinuity	X	
Cap not capable of being secured to bottle	X	
Metal caps not free of corrosion	X	
Bottle or cap not free of scratches, blemished or pits		X
Cap not clean		X
Bottle not free of imbedded foreign matter		X
Bottle not free of flash		X

4.2.1.2 Bottle dimensions examination. The bottle shall be examined for conformance to the dimensions specified in table I. Any nonconformance to the specified dimensions shall be classified as a defect. The lot size shall be expressed in units of bottles. The sample unit shall be one bottle. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.2.1.3 Box (for bottles) examination. The boxes shall be examined for the defects listed below. The lot size shall be expressed in units of boxes. The sample unit shall be one box. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for total (major and minor combined) defects.

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Defect	Classification Major	Minor
Partitions not complete or properly assembled in box	X	
Box not free of deformation	X	
Box not free of separation	X	
Box not free of tears, cuts, holes, splits, or punctures	X	
Box not capable of being properly closed	X	
Box or partitions not free of moisture	X	
Partition not free of deformation		X
Box or partition not free of dirt or foreign matter		X
Marking on box not complete, legible or present		X

* 4.2.2 Packaging inspection. An inspection shall be made to determine that assembly, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Material	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as: incomplete closure of container flaps, loose strapping or taping, inadequate stapling Bulged or distorted container
Content	Number of bottles per container not as specified Number of containers per shipping container is more or less than required Number of bottles, plastic bags, and sacks per shipping container not as specified (Type III only)

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4.3 Methods of inspection. The same bottles comprising the sample size (see 4.2.1) shall be used for all tests specified below. The test bottles shall not be included as part of the contract quantity.

* 4.3.1 Capacity test. The test shall be conducted at a room temperature of 77° ± 5°F. The bottles shall be filled with lubricating oil conforming to MIL-L-23699 to the transfer bead. The volume of oil used to fill the bottle shall be measured and recorded. Any bottle that does not meet the capacity requirement in 3.1.1 shall be a test failure.

4.3.2 Leakage test. The bottles shall be filled to the transfer bead with oil as specified in 4.3.1 and cap closure shall be applied hand tight. The bottles shall be tested for leakage by suspending them in an inverted position for 2 hours at room temperature and then suspended for 4 hours in an oven at 120° ± 10°F. Any evidence of leakage shall be a test failure.

* 4.3.3 Pen and pencil marking capability test. The letters and numbers "ABC-123" shall be printed with both ball point pen and pencil on the blocked-in ink areas of the bottles. Lack of adhesion of the pen and pencil marking to the white ink shall be a test failure.

* 4.3.4 Legibility after marking test. The ball point pen marking as performed in 4.3.3 shall be given one firm thumb rub. Illegibility of any letter or number shall be a test failure

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.1.1 Level A packing.

5.1.1.1 Type I. Ten oil sample container assemblies, assembled as specified in 3.2.2, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2.

5.1.1.2 Type II. Sixty mailing kit container assemblies, assembled as specified in 3.2.3, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. They shall be arranged ten in length, two in width, and three in depth with approximate inside dimensions of 20 inches in length, 10 inches in width and 9 inches in depth. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2.

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5.1.1.3 Type III. Twenty-four capped bottles shall be placed upright on the bottom of the shipping container. Two stacks (12 each) of paper sacks shall be placed directly on the bottles. Two stacks (12 each) of plastic bags shall be placed directly on the paper sacks. The shipping container shall be a snug-fitting fiberboard container conforming to style RSC, grade V2s of PPP-B-636. The approximate inside dimensions of the shipping container shall be 12 inches in length, 10-1/4 inches in width, and 6 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2.

5.1.2 Level B packing.

5.1.2.1 Type I. Ten oil sample container assemblies, assembled as specified in 3.2.2, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2.

5.1.2.2 Type II. Sixty mailing kit container assemblies, assembled as specified in 3.2.2, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. They shall be arranged ten in length, two in width, and three in depth with approximate inside dimensions of 20 inches in length, 10 inches in width and 9 inches in depth. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2.

5.1.2.3 Type III. Twenty-four capped bottles shall be placed upright on the bottom of the shipping container. Two stacks (12 each) of paper sacks shall be placed directly on the bottles. Two stacks (12 each) of plastic bags shall be placed directly on the paper sacks. The shipping container shall be a snug-fitting fiberboard container conforming to style RSC, grade W6c of PPP-B-636. The approximate inside dimensions of the shipping container shall be 12 inches in length, 10-1/4 inches in width, and 6 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2.

5.1.2.4 Weather-resistant shipping containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s box fabricated in accordance with PPP-B-636 and closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2.

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5.1.3 Commercial packing. Ten type I container assemblies, 60 type II container assemblies, or twenty-four type III (bottles, bags and sacks) shall be packed in accordance with ASTM D 3951.

5.2 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

* 6.1 Intended use. The oil sample container assemblies are used for taking oil samples from surface vehicles and mailing to laboratories for analysis. The oil sample bottle is designed to screw onto the "vampire" oil sampling pump.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this document.

(b) Type required (see 1.2).

(c) Selection of applicable levels of packing (see 5.1).

(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.1.2.4).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.1).

6.4 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

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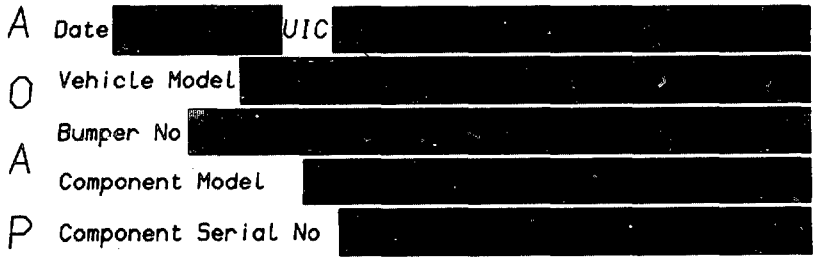


FIGURE 1. OIL SAMPLE BOTTLE MARKING

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44054B	BOTTLE: OIL SAMPLE; WITH CONTAINER, NONAERONAUTICAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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